

Salivary Diagnostics

Biochemical Markers for Systemic Diseases

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Introduction

- Saliva is an important diagnostic fluid.
- Contains numerous biochemical substances similar to blood.
- Used for detection and monitoring of systemic diseases.
- Emerging tool in modern biochemistry and molecular diagnostics.

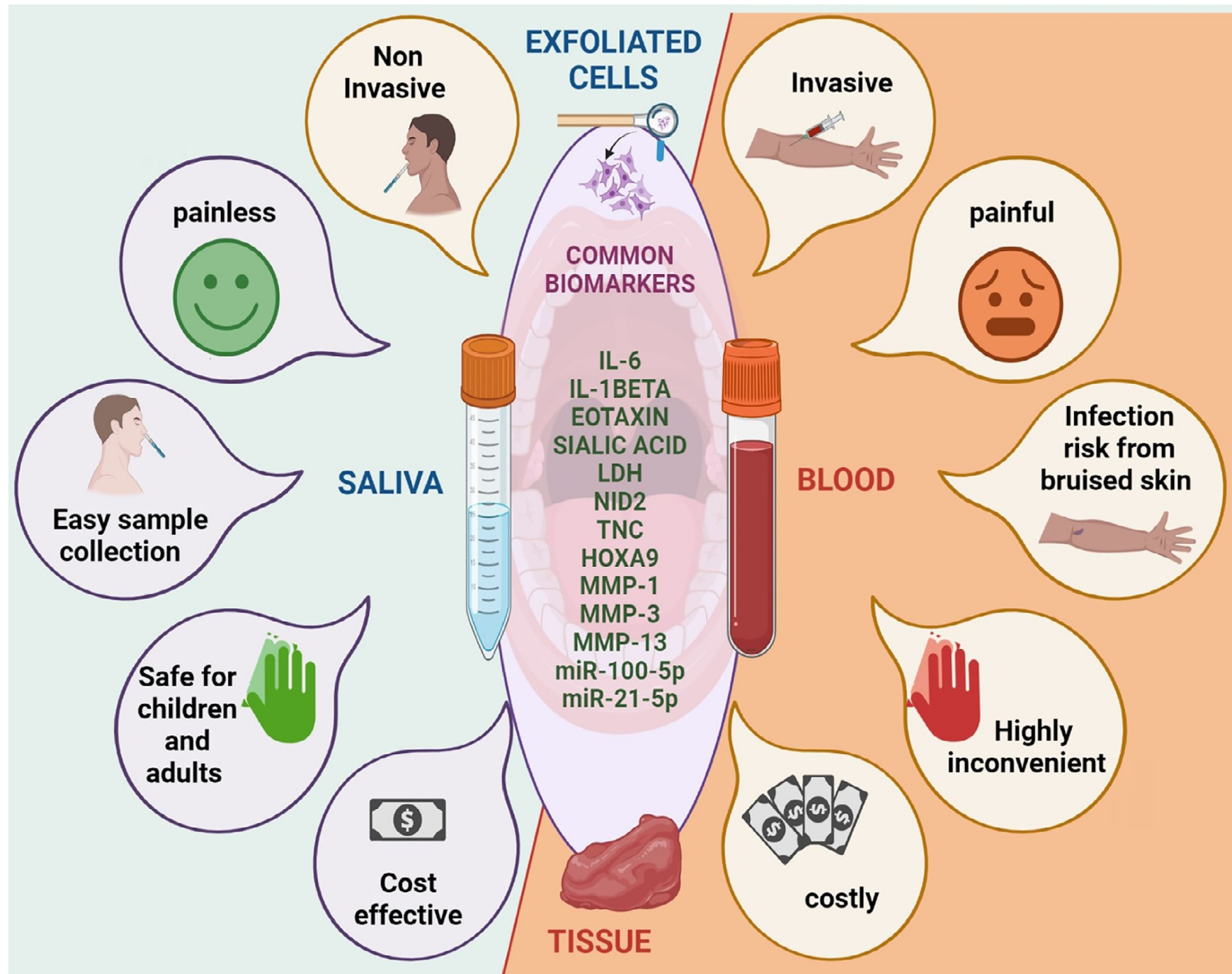
Definition

Salivary diagnostics refers to the use of saliva to identify biomarkers associated with health and disease.

Why Saliva?

❖ **Advantages of Salivary Diagnostics**

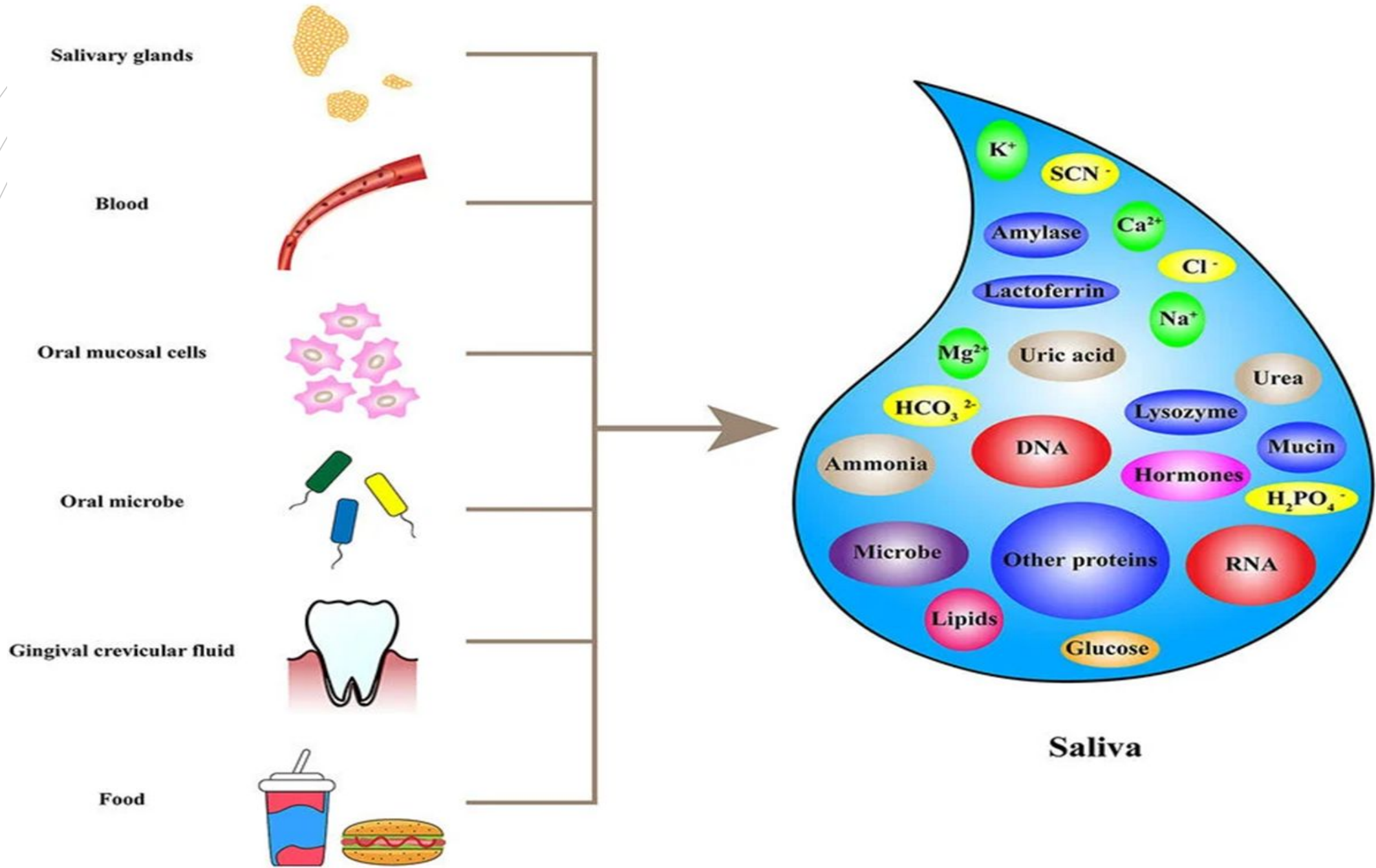
- Non-invasive
- Painless sample collection
- Easy handling and storage
- Low risk of infection
- Cost-effective
- Suitable for frequent monitoring



Composition of Saliva

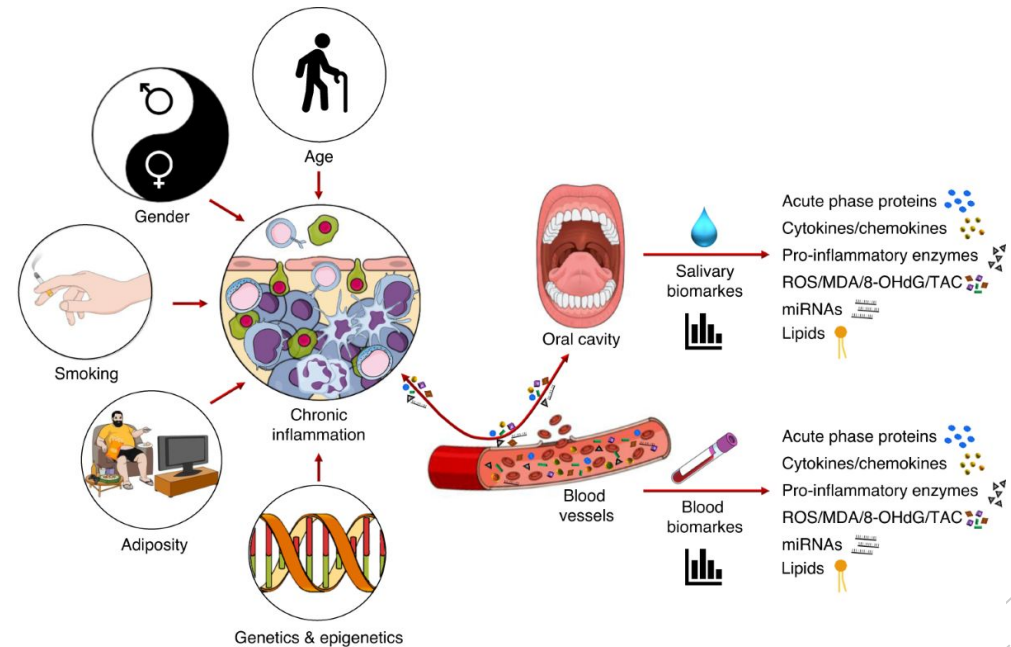
- **Major Constituents**
- Water (99%)
- Electrolytes
 - Na^+
 - K^+
 - Ca^{2+}
 - HCO_3^-
- Proteins & enzymes
- Hormones
- Antibodies
- DNA/RNA molecules

Source



Formation of Salivary Biomarkers

- Biomolecules enter saliva through:
 - Passive diffusion
 - Active transport
 - Ultrafiltration
- **Important Concept**
- Systemic biochemical changes may be reflected in saliva.



Types of Salivary Biomarkers

Biomarker	Examples	Clinical Use
Proteins	CRP, cytokines	Inflammation
Enzymes	LDH, amylase	Tissue damage
Hormones	Cortisol	Stress assessment
Antibodies	HIV antibodies	Infection diagnosis
Nucleic acids	miRNA, DNA	Cancer detection
Metabolites	Glucose, urea	Diabetes/renal disease

Diabetes Mellitus

❖ Salivary Biomarkers

- Glucose
- Amylase
- Cortisol

❖ Clinical Importance

- Salivary glucose correlates with blood glucose levels.
- Useful for non-invasive diabetic monitoring.

Cardiovascular Diseases

❖ **Important Biomarkers**

- C-reactive protein (CRP)
- Troponin
- Myoglobin

❖ **Significance**

- Early detection of myocardial infarction.
- Monitoring inflammatory status.

Cancer Diagnostics

◆ **Salivary Cancer Biomarkers**

- IL-8
- TNF- α
- p53 mutations
- microRNAs

◆ **Breast/Pancreatic Cancer**

- Specific mRNA and microRNA markers

◆ **Applications**

- Oral cancer screening
- Early diagnosis
- Prognostic evaluation

Infectious Diseases

◆ **Detectable Pathogens**

- HIV
- Hepatitis viruses
- SARS-CoV-2

◆ **Advantages**

- Rapid chairside testing and screening
- Point-of-care testing
- Reduced exposure to blood-borne infection

Autoimmune & Renal Diseases

◆ Autoimmune Disorders

○ Sjögren Syndrome

- Autoantibodies
- Cytokines

Significance

- Salivary gland dysfunction assessment.

◆ Renal Disease Biomarkers

- Urea
- Creatinine
- Uric acid

Significance

- Reflect renal filtration status.

Hormonal & Stress Markers

◆ **Salivary Hormones**

- Cortisol
- Testosterone
- Melatonin

◆ **Uses**

- Stress analysis
- Endocrine disorders
- Circadian rhythm studies

Diagnostic Techniques

◆ **Conventional Methods**

- ELISA
- PCR
- Spectrophotometry

◆ **Advanced Technologies**

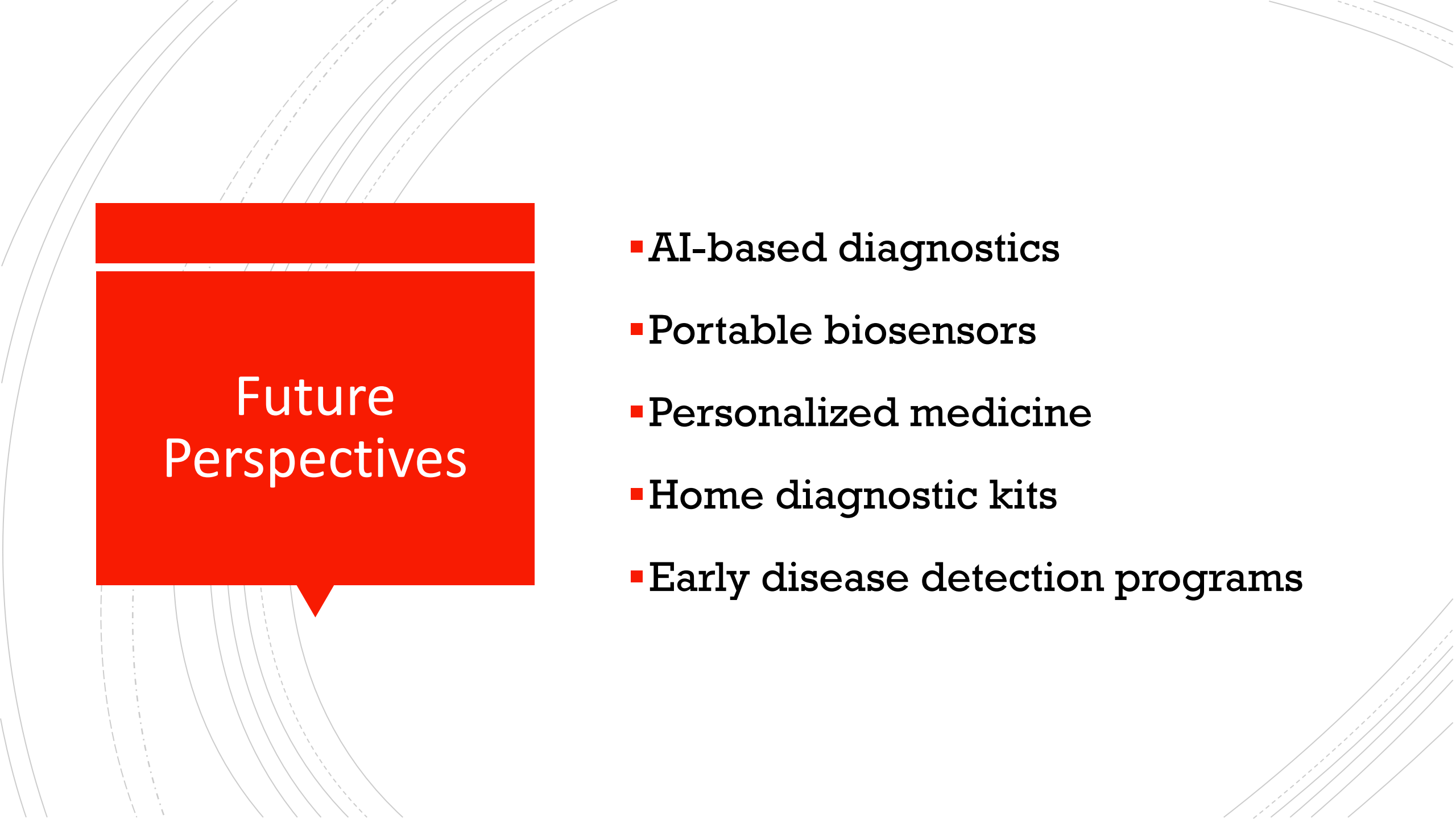
- Biosensors
- Nanotechnology
- Lab-on-a-chip systems
- Microfluidics

Advantages

- Non-invasive sampling
- Patient compliance
- Safe and convenient
- Useful for mass screening
- Real-time monitoring potential
- Low cost
- Minimal training required
- Easy repeated sampling

Limitations

- Low biomarker concentration
- Variability due to:
 - Flow rate
 - Diet
 - Medications
 - Circadian rhythm
- Contamination risks
- Lack of standardization
- Need for standardized collection methods
- Sensitivity concerns



Future Perspectives

- AI-based diagnostics
- Portable biosensors
- Personalized medicine
- Home diagnostic kits
- Early disease detection programs

Conclusion

- Salivary diagnostics is an emerging and promising field in modern medicine and dentistry.
- Saliva is a promising diagnostic fluid.
- Reflects systemic biochemical alterations.
- Advances in biotechnology are enhancing diagnostic accuracy.
- It has great potential for early diagnosis, disease monitoring, and personalized healthcare.
- Salivary diagnostics may become a routine clinical tool in future healthcare.



“Thank you”